# RAG (Retrieval-Augmented Generation)
## Definition

RAG (Retrieval-Augmented Generation) is an AI architecture that combines information retrieval with Large Language Models (LLMs). Instead of relying only on the model's trained knowledge, RAG retrieves relevant information from external sources and provides it to the LLM before generating a response.

## Why Do We Need RAG?

LLMs have some limitations:

Knowledge can become outdated.
They may generate incorrect information (hallucinations).
They do not have access to private company data by default.

RAG solves these problems by retrieving the latest and most relevant information from external data sources before generating an answer.

## How RAG Works
### Step 1: User Query

The user asks a question.

Example:

What are the authentication methods used in our project?
### Step 2: Retrieval

The system searches documents, databases, or vector stores to find relevant information.

### Step 3: Context Injection

The retrieved information is added to the prompt.

### Step 4: Generation

The LLM uses the retrieved context to generate an accurate response.

## RAG Architecture
```
User Question
     ↓
Retriever
     ↓
Relevant Documents
     ↓
Prompt + Context
     ↓
LLM
     ↓
Generated Answer
```

## Main Components of RAG

### 1. Data Source

Contains the knowledge to be searched.

Examples:

PDFs
Word Documents
Databases
Websites
Company Knowledge Bases

### 2. Embedding Model

Converts text into numerical vectors.

Purpose:

Enables semantic search.
Finds meaning-based matches instead of keyword matches.

### 3. Vector Database

Stores embeddings for efficient retrieval.

Examples:

Chroma
FAISS
Elasticsearch

### 4. Retriever

Searches the vector database and returns the most relevant chunks.

### 5. LLM

Generates the final response using the retrieved context.

## Advantages of RAG

Provides more accurate answers.
Reduces hallucinations.
Uses up-to-date information.
Can access private organizational data.
No need to retrain the model whenever documents change.

## Disadvantages of RAG

Retrieval quality affects response quality.
Additional infrastructure is required.
Slightly higher response time due to retrieval step.

## Real-World Example

Suppose your company has a PDF containing project documentation.

Without RAG:

User → LLM → Answer

The LLM may not know the project details.

With RAG:

User
↓
Retriever
↓
Project Documents
↓
LLM
↓
Accurate Answer

The answer is generated using information from the project documents.

RAG in Your AI Knowledge Management System
User Question
↓
FastAPI API
↓
Elasticsearch Retrieval
↓
Relevant Chunks
↓
LangChain
↓
LLM (Ollama/Llama)
↓
Final Response

Here:

Elasticsearch stores and searches document chunks.
LangChain manages retrieval and prompt creation.
The LLM generates the final answer.
Interview Answer (1 Minute)

"RAG, or Retrieval-Augmented Generation, is an AI architecture that combines information retrieval with Large Language Models. When a user asks a question, the system first retrieves relevant information from external sources such as documents, databases, or vector stores. This retrieved context is then provided to the LLM, which generates a response based on both the retrieved information and its existing knowledge. RAG improves accuracy, reduces hallucinations, and enables AI systems to answer questions using up-to-date or private organizational data."

One-Line Definition

RAG is a technique that enhances an LLM by retrieving relevant information from external knowledge sources before generating a response.